

M54 — Collapsible threshold panel + concurrent tiered repeating audio + magnitude-based threshold severity

Phase: Phase 7 follow-on (post-M53, operator feedback) **Status:** **DONE** **Blocked by:** M48 (thresholds), M49 (sounds), M50 (severity tiers), M52 (repeating audio) **Blocks:** none **Estimated effort:** 0.5 day

1. Purpose

"Nursing station- alarm threshold click to drop down and open, click header of Alarm threshold and have it roll up. Alarms need to run concurrently. For higher priority alarms (low- medium-High) the alarms sound more frequently. For example the sound loop for a code blue should run continuously with minimal time gap running the sound loop. Low level alarms sound when parameters are 10% below lower threshold or 10% Higher than upper threshold. Medium is between 10% and 20% of threshold limits and high if above 20% of the threshold limits"

Four asks bundled:

1. **Collapse / expand the threshold panel** by clicking its header. 2. **Concurrent audio** — multiple alarms play simultaneously. 3. **Tighter, tiered cadence** — code blue near-continuous; lower tiers progressively less frequent. 4. **Magnitude-based threshold severity** — 0–10% past bound is LOW (info), 10–20% is MEDIUM (warning), >20% is HIGH (critical).

2. What ships

2.1 Collapsible alarm-threshold panel

portal/templates/nurse_station.html — the threshold <section> default-collapses with class ns-collapsed; the <h2> becomes a role=button toggle (#ns-thresholds-toggle) with aria-expanded + aria-controls. A caret span rotates 90° when expanded.

portal/static/nurse_station.js — new wireThresholdToggle():

- Click on the H2 → toggles ns-collapsed on the section, flips aria-expanded.

- Keyboard accessible: Enter / Space when focused also toggles.

portal/static/nurse_station.css — .ns-thresholds.ns-collapsed .ns-thresholds-form { display: none; } hides the form; the header stays visible. Caret rotates via a CSS transition.

2.2 Audio cadence — four tiers

New table in nurse_station.js:

Tier	Cadence	Maps to severity
danger	2500 ms	danger (code blue, dangerous rhythm) — near-continuous
high	5000 ms	critical
medium	15000 ms	warning
low	35000 ms	info

Pre-M54: high=8000, medium=20000, low=45000 — tightened across the board.

The dispatcher's tier-picker reads severity first (so danger gets its own bucket even though severity_to_priority maps both critical and danger to "high" for the WAV lookup — same WAV file, different cadence):

```
function _audioCadenceTier(a) {
  const sev = (a.severity || '').toLowerCase();
  if (sev === 'danger') return 'danger';
  return a.audio_priority || 'medium';
}
```

2.3 Fast ticker for the danger tier

The state poll runs every 3 seconds — that's too slow for the 2.5 s danger cadence. M54 adds a 700 ms `setInterval` that re-runs the dispatcher against the cached last-alarms list:

```
let _lastAlarmsForAudio = [];
// pollOnce caches alarms here on every refresh.
setInterval(() => {
  if (_lastAlarmsForAudio && _lastAlarmsForAudio.length) {
    _playNewAlarmSounds(_lastAlarmsForAudio);
  }
}, 700);
```

For non-danger tiers (≥ 5 s cadence) the 3 s poll already gates the re-fires — the fast ticker is effectively a no-op for them. The 2.5 s danger tier is the one that benefits.

2.4 Concurrent playback

The dispatcher iterates `alarms.forEach(a => { ... new Audio(url).play(); })` — each iteration creates an independent `Audio` instance, so two beds in simultaneous breach => two overlapping tones. M54 adds an explicit `// CONCURRENTLY` comment so future editors don't accidentally serialise into a queue.

2.5 PIA cascade cadence tightened

`portal/static/pia_app.js` — `CASCADE_AUDIO_REPEAT_MS 8000 → 2500`. Same intent: code blue near-continuous on the bedside tablet.

2.6 Magnitude-based threshold severity

`portal/alarms.py::_threshold_alarms_for` now computes `deviation_pct = (value - bound) / abs(bound) * 100` and maps:

```
if deviation_pct >= 20.0:
    severity = "critical"      # HIGH tier
elif deviation_pct >= 10.0:
    severity = "warning"      # MEDIUM tier
else:
    severity = "info"         # LOW tier
```

Pre-M54: per-metric fixed severity (SpO2 always critical, others warning). Now responsive to the depth of the breach — a 1-point SpO2 dip below threshold is now `info`, a 22%+ dip stays `critical`.

Each threshold alarm dict gains a `deviation_pct: float` field so the UI could render a "+15% over upper" badge in the future.

2.7 Code-blue / code_blue_button promoted to danger

`portal/alarms.py::_SEVERITY_BY_KIND`:

- "code.blue": "danger" (was "critical")
- "code_blue_button": "danger" (was "critical")

This puts code blue at the top of the sort order AND routes its audio through the new 2.5 s danger cadence. Same WAV.

3. Files touched

Modified

- portal/alarms.py — `_SEVERITY_BY_KIND` code.blue + code_blue_button → "danger"; `_threshold_alarms_for` computes magnitude-based severity + adds `deviation_pct` to each alarm.
- portal/templates/nurse_station.html — `<section class="ns-thresholds ns-collapsed">` + `role=button` H2 toggle with ARIA.
- portal/static/nurse_station.js — new `wireThresholdToggle()`, four-tier `AUDIO_REPEAT_MS`, `_audioCadenceTier()` helper, 700 ms fast ticker, `_lastAlarmsForAudio` cache, M54 concurrency comment.
- portal/static/nurse_station.css — `.ns-thresholds-toggle` + `caret` + `.ns-collapsed` rules.
- portal/static/pia_app.js — `CASCADE_AUDIO_REPEAT_MS` 8000 → 2500.

New

- tests/v7/test_collapse_concurrent_tiered_severity.py — 17 acceptance tests.
- docs/module_guides/M54_collapsible_concurrent_tiered_severity.{md, pdf} — this guide.

Tests updated (intentional contract changes)

- tests/v7/test_alarm_bus.py — code.blue severity → danger.
- tests/v7/test_alarm_thresholds_and_ecg_fix.py — HR=130 against high=80 → critical (was warning, since 62.5% > 20%); SpO2 deep-drop test deepened to value=70 so it stays in critical under the new magnitude rules.
- tests/v7/test_clinical_alarm_sounds.py — SpO2 audio-priority test deepens to 70 to land in the high-priority bucket; cadence regex search window widened to accommodate the larger dispatcher.
- tests/v7/test_future_devices.py — code_blue_button severity → danger.
- tests/v7/test_repeat_audio_silence_brand.py — PIA cascade cadence now 2500 (was 8000).

4. Acceptance

Source: tests/v7/test_collapse_concurrent_tiered_severity.py (17 tests).

4.1 Magnitude-based severity

- HR.high=100, v=105 → info, deviation_pct=5.
- HR.high=100, v=115 → warning, deviation_pct=15.
- HR.high=100, v=125 → critical, deviation_pct=25.
- HR.low=60, v=42 → critical (30% below).

- `SpO2.low=90, v=70` → critical (22% below).
- `SpO2.low=90, v=86` → info (~4% below).

4.2 Code blue at top tier

- `code.blue` scene alarm has `severity="danger"`.
- `code_blue_button` (M29) `severity="danger"`.
- Code-blue WAV unchanged — still `03_code_blue.wav`.

4.3 Tiered + concurrent audio

- `AUDIO_REPEAT_MS` table has all four keys with 2500/5000/15000/35000.
- `_audioCadenceTier` reads `a.severity` and routes "danger" to its own tier.
- 700 ms fast ticker exists with `_lastAlarmsForAudio` cache.
- Dispatcher creates one new `Audio(url)` per alarm + a `CONCURRENTLY` comment.

4.4 PIA cascade tightened

- `CASCADE_AUDIO_REPEAT_MS` = 2500 (old 8000 constant removed).

4.5 Collapsible panel

- Section renders with class `ns-thresholds ns-collapsed` on first paint.
- Toggle element has id `ns-thresholds-toggle` with `aria-expanded="false"` + `aria-controls="ns-thresholds-form"`.
- Caret span present.
- JS has `wireThresholdToggle`, `ns-collapsed` class manipulation, Enter/Space keyboard handling.
- CSS hides the form when collapsed; caret rotates when expanded.

All 17 pass. Full v7 suite **441 passed, 1 skipped, 0 regressions** (was 424 pre-M54). Five pre-existing tests updated to reflect M54's intentional contract changes (code-blue severity, magnitude-based threshold severity, PIA cascade cadence).

5. Cadence math — why these numbers

Real bedside-monitor cadences are roughly:

- IEC 60601-1-8 high-priority alarms: 2.5–5 second repeat interval.
- Medium: 10–30 seconds.
- Low: ≥ 30 seconds.

M54 lands on:

- danger 2.5 s — operator wants code blue "with minimal time gap".

2.5 s is the IEC lower bound and the shortest cadence that doesn't bury other concurrent alarms.

- high 5 s — at the spec's upper bound for high-priority audio.
- medium 15 s — middle of the IEC range.

- low 35 s — long enough to avoid ear fatigue for low-acuity alarms

(a slipped call-bell shouldn't sound every 8 seconds).

For the magnitude→severity thresholds (10% / 20%), the operator's phrasing is taken literally:

- 0–10% past bound = LOW
- 10–20% past bound = MEDIUM
- > 20% past bound = HIGH

This matches typical clinical practice: a small dip across the boundary is a hint, a deep drop is an emergency.

6. Operator demo

1. Open the Nursing Station — the **Alarm thresholds** section is collapsed by default; click the header to expand. Click again to collapse. The caret rotates. 2. From a Per-Patient Console, set SpO2 = 88 (~2% below the 90 default threshold). The Nursing Station alarm fires at **info** severity (greyed-out row) with a low chime every ~35 s. 3. Drop SpO2 to 80 — alarm escalates to **warning** (yellow), chime every ~15 s. 4. Drop SpO2 to 65 — alarm escalates to **critical** (red), chime every ~5 s. 5. Inject a code . blue scene from any encounter — top-row red alarm with **danger** severity; the chime now sounds every ~2.5 s, near-continuous. 6. With code blue still active, inject a second alarm (HR or SpO2). Both tones play simultaneously — code blue's repeating high-pri tone over the other tier's chime. 7. Silence the code blue — both stop. After 45 s the silence expires; if still active, audio resumes at the same cadence.

7. What was deliberately NOT changed

- **WAV library** — `_METRIC_FILES` and `_SPECIAL_FILES` are

unchanged. `severity_to_priority` still maps `danger` → `high` so the WAV lookup keeps working with the existing assets. The new `danger` cadence tier is JS-side only.

- **Threshold-alarm sort order** — threshold alarms still use

`ts=0` so they sort to the BOTTOM of their severity bucket. A critical-severity threshold breach sorts below a critical-severity device alarm with a real timestamp. M48 invariant preserved.

- **PIA flash animation** — gated by `_cascadeKey` (NEW alarm set

triggers a restart), not by audio cadence. Don't want the CSS animation restarting every 2.5 s; the steady flash that runs while `pia-cascade-active` is sufficient.

- **No persistence** — collapsed/expanded state of the threshold

panel is in-memory only. Reload resets to collapsed. If the operator wants the preference to persist, add a `localStorage` read/write in `wireThresholdToggle`.

Closes: all four M54 asks in full.